sltoo – Integrating Requirements into CI/CD

Kristoffer Nordström

September 1, 2021

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  - More pictures
- Contribution of mine hervorstreichen
- Mehr Beispiele
  - Personify Eugen from Marketing (?)

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Motivation

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It's an honour to open this topic's session.

When talking about requirements, we should also define the requirements of the problem we're trying to solve. For all requirements there's a anecdote.

Storytime

- Fully automated traceability matrix
- Consistent and up-to-date documents from the source
- Store (requirements') meta-information with code

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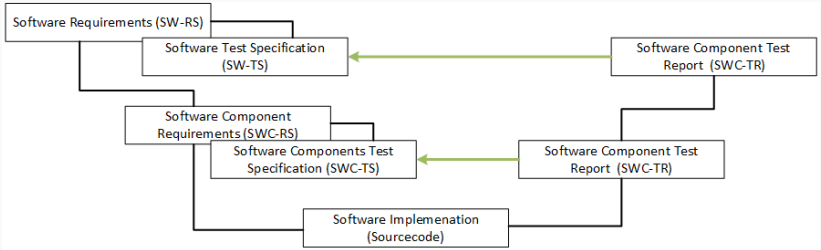
- It's fine manually. Requirements change find minimal delta-test? Change more than once. Hence we want this fully automated.
- Review multiple documents for a different contractor. No consistent set of documents were provided
- Jon Holt's term in introduction to MBSE: Documents are a live *view* of the system (not pretty pictures). CI/CD for software, why not for documents?
- It came with one document that contained all requirements and use-cases. It appeared to have been exported from some web-based tool. All the information therein is lost
- Also: Too technical, interface with management/business side
- 4'

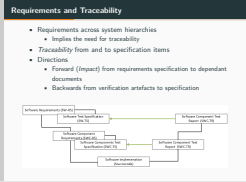
Theory

Theory

Requirements and Traceability

- Requirements across system hierarchies
  - Implies the need for traceability
- *Traceability* from and to specification items
- Directions
  - Forward (*Impact*) from requirements specification to dependant documents
  - Backwards from verification artefacts to specification





Here Traceability only from items/issues

Traceability

- *Requirement A*
  - Red button to shut down system
- *Implementation a* says implemented A
  - Traceability can be automated
  - Machine-readable
- What if *A* changes?
  - *A* knows nothing of *a*

Traceability
Requirement A
Red button to shut down system
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A knows nothing of a
PANIC

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DON'T PANIC

Proposed Solution

- *Requirement A-1.0* says do `A`
- *Implementation a* says implemented `A-1.0`
  - *A-1.0* changes to *A-2.0* do `Ã`
  - Use hashes instead of semantic versioning
  - Calculated automatically

DON'T PANIC

Proposed Solution
Requirement A-1.0 says do A
Implementation a says implemented A-1.0
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Use hashes instead of semantic versioning
Calculated automatically
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Example Requirement

VCD Writer Inputs SW-AS-501
The output from *AggregatorInterface* can be fed into the *VCD Writer* to create an identical output file.
Rationale: Make the process as transparent as possible.
Depends on: SW-AS-500

Id:	SW-AS-501	Priority:	0.0	Owner:	development
Invented on:	2020-05-30	Invented by:	default	Status:	passed
Class:	detailable	Hash:	f8d68d11		

Hash is calculated over *Name*, *Description* and *Verification Method*.
Rationale is only informative

```
$ sha256sum "${Name}${Description}${VerifMethod}"
```


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Testing the Example Requirement

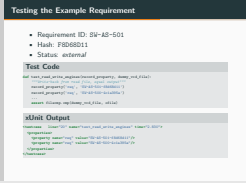
- Requirement ID: `SW-AS-501`
- Hash: `F8D68D11`
- Status: *external*

Test Code

```python
def test_read_write_engines(record_property, dummy_vcd_file):
    """Write-back from read file, equal output"""
    record_property('req', 'SW-AS-501-f8d68d11')
    record_property('req', 'SW-AS-500-4c1a395a')
    ...
    assert filecmp.cmp(dummy_vcd_file, ofile)
```

xUnit Output

```xml
<testcase   line="20" name="test_read_write_engines" time="2.830">
  <properties>
    <property name="req" value="SW-AS-501-f8d68d11"/>
    <property name="req" value="SW-AS-500-4c1a395a"/>
  </properties>
</testcase>
```

Traceability Matrix

Traceability Matrix

Table A.1: Traceability Matrix Table

Req. ID	Status	UT
SW-AS-1	passed	passed
SW-AS-100	open	open
SW-AS-101	open	open
SW-AS-102	open	open
SW-AS-200	passed	passed
SW-AS-201	passed	passed
SW-AS-202	passed	passed
SW-AS-300	passed	passed
SW-AS-301	open	open
SW-AS-302	open	open
SW-AS-500	passed	passed
SW-AS-501	passed	passed

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SW-AS-500 passed passed
SW-AS-501 passed passed

Integrating Requirements into CI/CD

- Integration for every output document
- Match *open* and/or *failed* issues
- Example for *failed* issues

```
$ bash -ec 'test "$(grep -c failed \
    arch/artifacts/tracematrix.tex)" -eq "0"'
```


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$ bash -ec 'test "$(grep -c failed \
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Document Baseline

Every document has a its own version tag

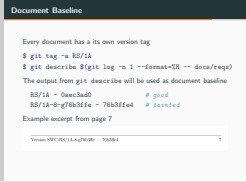
```
$ git tag -a RS/1A
$ git describe $(git log -n 1 --format=%H -- docs/reqs)
```

The output from `git describe` will be used as document baseline

```
RS/1A - 0aec3ad0              # good
RS/1A-8-g76b3ffe - 76b3ffe4   # tainted
```

Example excerpt from page 7

Version SWC-RS/1A-8-g76b3ffe — 76b3ffe4 7



This is all nice and dandy; but how do we communicate with people for whom git clone is too much to ask?

Business Interface

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Excel Workflow

- Non-tech people define the requirements
- Familiarity / Ease-of-use
- Consistency of Documents
  - Works if all you've got is Office and E-Mail
- INSERT IMAGE HERE

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- Requirements shipped with code ✓
  - Including relational meta-information
- Traceability matrix automated ✓
- Continuously updated documentation ✓
  - Document Versioning (baselining) ✓

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Your Jira workspace will be gone

Questions

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Appendix

Excel Support Rationale

- Good-enough GUI
- Understood by everyone
- The *Truth* is still in your repository
  - Import from Excel to repository
  - Export every build to a new Excel file
  - Templating for branding

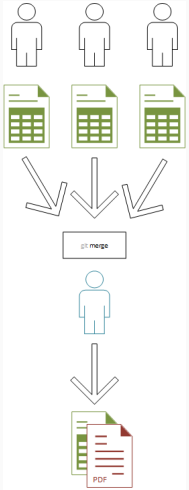

Figure 1: Workflow

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Good-enough GUI
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rmtoo – Introductions

An introduction presentation into *rmtoo* and with more details.

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- Traceability for the given requirements
- Bring code and documentation into same repository
- Integrate into build-system
  - Detect upstream changes to requirements
  - Quickly identify affected code-regions
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Results

The status *external* will yield the following results:

- *open*
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- *passed*
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  - Unit-tests passed
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Installation

Traceability features are in the beta releases.

```
$ pip3 install sltoo>=25.1.0b3
$ wget https://kown7.github.io/pymergevcd/assets/template_p
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Future Developements

- Write Parser for *Test Reports* ✓
- Documents with the correct identifier automatically solve the specification
  - Document Formats:
    - docx (maybe with pandoc)
    - LaTeX✓
    - Text

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Final Thoughts

- Never test against your requirements
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Baseline

- Based on rmtoo
  - File-based requirements tracking tool
  - Requirements stored in text-files
  - group requirements by topics
  - document and requirements LaTeXbased
    - Customizable with templates
- Manage with your favourite VCS
  - Merge changes from various sources
- Excel import/export
- Traceability

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```
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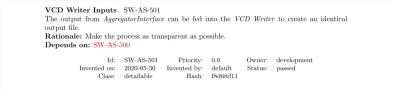
Status: external
Owner: development
Effort estimation: 1
Invented on: 2020-05-30
Invented by: default
Type: requirement
```



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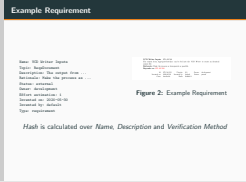
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